SPACE

Hubble Network plans massive satellite upgrade to create global Bluetooth layer

Aria Alamalhodaei — 4:00 AM PDT · August 7, 2025

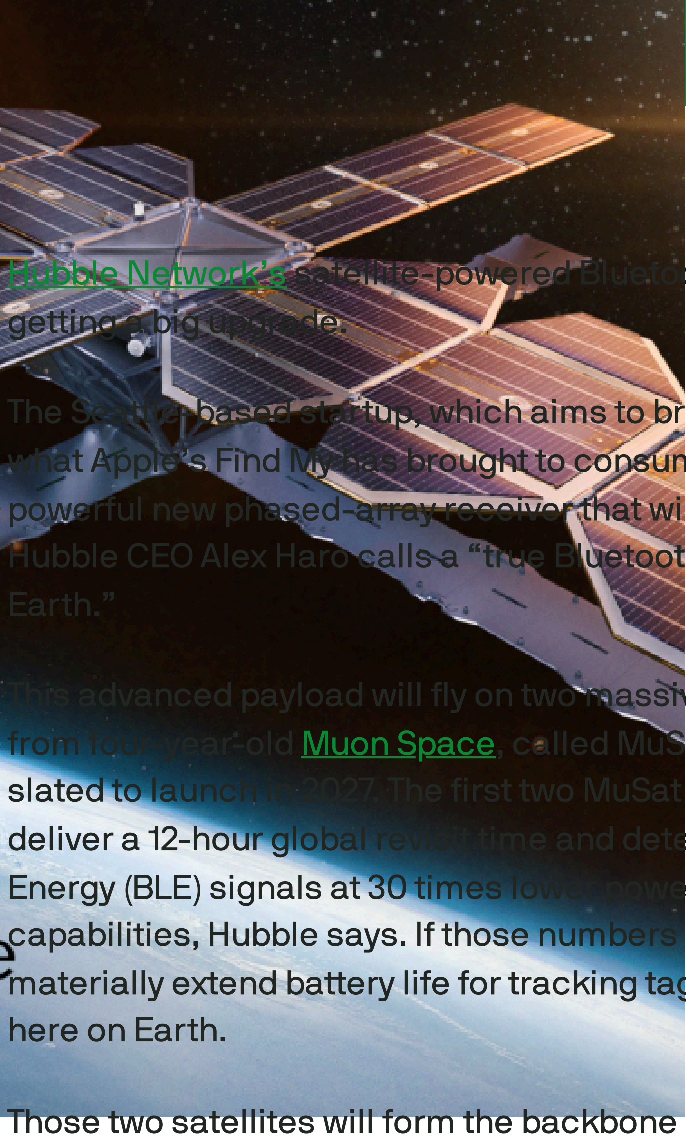

IMAGE CREDITS: MUON SPACE

Hubble Network's satellite-powered Bluetooth network is getting a big upgrade.

The Seattle-based startup, which aims to bring to enterprises what Apple's Find My has brought to consumers, has built a powerful new phased-array receiver that will enable what Hubble CEO Alex Haro calls a "true Bluetooth layer around the Earth."

This advanced payload will fly on two massive new satellites from four-year-old Muon Space, called MuSat XL, which are slated to launch in 2027. The first two MuSat XL spacecraft will deliver a 12-hour global revisit time and detect Bluetooth Low Energy (BLE) signals at 30 times lower power than current capabilities, Hubble says. If those numbers pan out, it could materially extend battery life for tracking tags and sensors here on Earth.

Those two satellites will form the backbone of Hubble's BLE Finding Network for enterprises in sectors ranging from logistics, infrastructure, and defense.

Hubble made history in 2024 when it became the first company to establish a Bluetooth connection directly to a satellite. The startup's proposition is compelling: instead of needing to buy specialized hardware, customers will only need to integrate their devices' chipsets with a piece of firmware to enable connection to the Hubble network.

The benefits of the space-based network are massive, Hubble argues: it can provide visibility across the globe, including in remote areas, and offers a developer-friendly way to let companies track assets without building any additional infrastructure.

The company currently has seven spacecraft on orbit with a target of having 60 satellites in operation by 2028. The long-term goal is to upgrade the entire constellation to the larger platform buses because of their power and performance upgrades, Haro said.

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It's an aggressive schedule, but Haro added that one reason Hubble chose to partner with Muon is the latter's ability to rapidly scale manufacturing to hit this goal, despite being a young company. (A recent $146 million funding round should help.) Gregory Smirin, president of Muon Space, told TechCrunch that the company's San Jose production facility is being built out to support production of over 500 spacecraft per year by 2027.

Hubble is the first customer for the 500 kilogram-class MuSat XL satellite platform, which Muon says can provide multi-kilowatt power to payloads, optical crosslinks, high-volume downlink, and "near real-time" communications for time-sensitive missions. The partnership also signals a bigger push by the company to compete for more lucrative contracts with the Department of Defense.

The XL platform "is a perfect size and capability for SDA Tranche missions," Smirin said, referring to the Space Development Agency's initiative to build out a missile defense constellation in low Earth orbit. "XL reflects both the evolution of our technical stack and our growing role in delivering the kind of multi-mission spacecraft that programs like SDA increasingly rely on."

Muon's business model can be thought of as space-as-a-service: the company designs, builds, and operates satellites using a vertically-integrated stack of hardware and software. The stack, called Halo, is meant to open up space access for companies with promising payloads but no interest in building the underlying satellite architecture. In practice, that means Hubble can focus on developing the BLE network while Muon handles the satellite platforms and mission operations.

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Muon Space closes $56M to scale all-in-one satellite platform

Aria Alamalhodaei — 12:36 PM PDT · August 5, 2024

Space-as-a-service startup Muon Space closed a $56.7 million fundraising round to scale its satellite platform, while also announcing a new deal with defense giant Sierra Nevada Corp.

California-based Muon designs, builds and operates satellites in low Earth orbit on behalf of customers. Some customers, like Hydrosat, provide their own payloads, while others, like the nonprofit coalition Earth Fire Alliance, use Muon-developed payloads. Muon's business model represents a new paradigm for access to space: Instead of spending years developing satellite hardware and flight software, mission operations, and data processing, customers can use Muon's end-to-end service instead.

The startup says it has secured over $100 million in committed contracts from customers this year alone for its Halo satellite platform. That includes $60 million in deals announced earlier this year to design 10 remote-sensing satellites for unnamed customers. More recently, the company also announced a new deal with Sierra Nevada for the development of three spacecraft for the Vindlér remote sensing constellation. The first Vindlér satellite built by Muon is scheduled to launch in 2025.

The Series B round was led by Activate Capital, with participation from Acme Capital and participating investors Costanoa Ventures, Radical Ventures and Congruent Ventures. It brings the company's total capital raised to over $91 million, including a 2022 Series A round and a seed round in 2021.

Topics: hydrosat muon space Sierra Nevada Corporation Space

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Hubble Network makes Bluetooth connection with a satellite for the first time

Aria Alamalhodaei — 1:00 PM PDT · May 2, 2024

Hubble Network has become the first company in history to establish a Bluetooth connection directly to a satellite — a critical technology validation for the company, potentially opening the door to connecting millions more devices anywhere in the world.

The Seattle-based startup launched its first two satellites to orbit on SpaceX's Transporter-10 rideshare mission in March; since that time, the company confirmed that it has received signals from the onboard 3.5mm Bluetooth chips from over 600 kilometers away.

The sky is truly the limit for space-enabled Bluetooth devices: The startup says its technology can be used in markets including logistics, cattle tracking, smart collars for pets, GPS watches for kids, car inventory, construction sites and soil temperature monitoring. Haro said the low-hanging fruit is those industries that are desperate for network coverage even once per day, like remote asset monitoring for the oil and gas industry. As the constellation scales, Hubble will turn its attention to sectors that may need more frequent updates, like soil monitoring, to continuous coverage use cases like fall monitoring for the elderly.

Once it's up and running, a customer would simply need to integrate their devices' chipsets with a piece of firmware to enable connection to Hubble's network.

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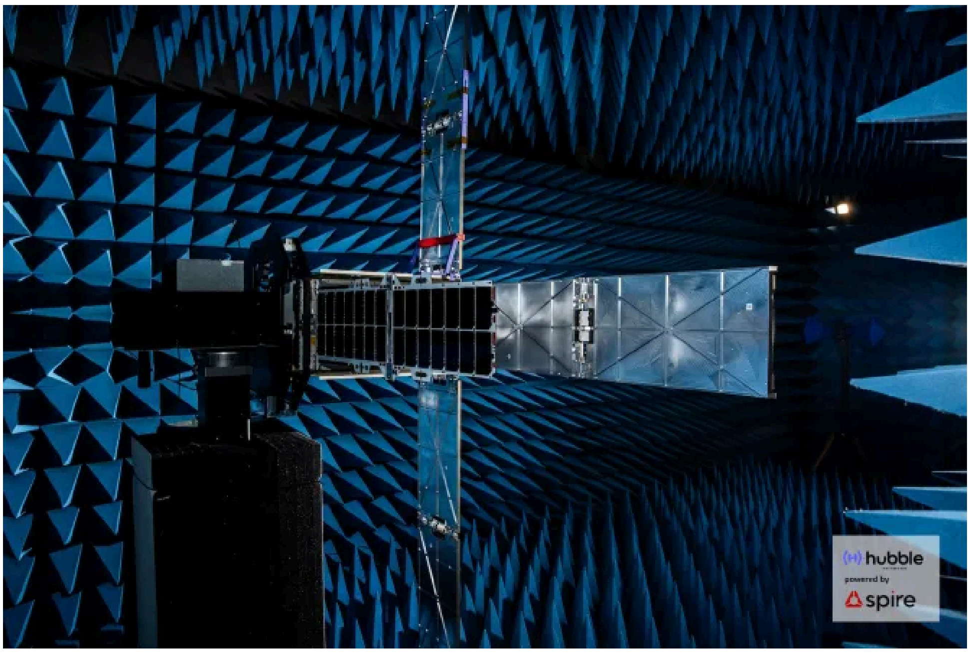
The company joined Y Combinator's Winter 2022 cohort and closed a $20 million Series A last March. Hubble's first innovation was to develop software enabling off-the-shelf Bluetooth chips to communicate over very long ranges with low power.

On the space side, the company also patented a phased array antenna that can launch on a small satellite. The antennas work almost like a magnifying glass, and it's what enables an off-the-shelf Bluetooth chip to communicate with the Hubble satellite. The team also had to solve Doppler-related problems, frequency mismatches that occur between fast-moving objects exchanging data via radio waves.



ONE OF HUBBLE'S SATELLITES IN A TERRESTRIAL TEST CHAMBER.

IMAGE CREDITS: HUBBLE NETWORK

Hubble is aiming to launch a third satellite on SpaceX's Transporter-11 mission this summer and a fourth on Transporter-13. Those four satellites will compose what Haro called the "beta constellation," and pilot customers are starting to turn their integrations on even today, he said. The startup plans to launch the following 32 satellites all at once in the fourth quarter of 2025 or the beginning of 2026, though the launch provider has not been selected yet.

Those 36 satellites will compose Hubble's first "production constellation," and they'll enable connection with a Hubble satellite roughly two-three hours per day from anywhere in the world.

Topics: Bluetooth hubble network Satellites Space Startups

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Hubble Network wants to connect a billion devices with space-based Bluetooth network

Aria Alamalhodaei — 6:00 AM PDT · May 31, 2023

Bluetooth-enabled devices are ubiquitous, but how those devices are used is constrained by the relatively short range provided by Bluetooth technology. Seattle-based startup Hubble Network wants to completely upend that status quo by launching a satellite network that any Bluetooth-enabled device can connect to, anywhere in the world.

IMAGE CREDITS: HUBBLE NETWORK

The company's aim is to build out a constellation of 300 satellites that can provide real-time updates for any sensor or device outfitted with a Bluetooth low energy (BLE) chip. On its website, Hubble proposes use cases that span industries — from child safety to pallet tracking to environmental monitoring. The startup's ultimate goal is to connect over a billion devices on its network.

Hubble Network CEO Alex Haro says the company has engineered "technical tricks" to make this scale of connectivity possible for the first time, like lowering the bitrate, or the amount of data transferred per second. Hubble has also rethought the design of the satellite antenna. Instead of sticking a single antenna on the side of a satellite bus, the company is using hundreds of antennae per satellite. This means that each satellite can support millions of connected devices.

"That is essentially a huge magnifying glass onto the surface of the earth that's able to detect these very weak radio signals coming out of the Bluetooth chips, and that's what enables you to actually decipher and receive the Bluetooth signal," Haro explained.

The result is a radio signal that can be detected around 1,000 kilometers away — or almost 10 orders of magnitude longer than what can be detected from a Bluetooth chip over terrestrial networks.

Hubble Network plans to launch an initial batch of four satellites on SpaceX's Transporter-10 rideshare mission in January 2024, and onboard early pilot customers after. The startup is fully funded through this mission, Haro said, thanks to a $20 million Series A round that closed in March. That round was led by Transpose Platform, with additional participation from 11.2 Capital, Y Combinator, Yes.VC, Convective Capital, Seraphim Space, Type One Ventures, Soma, AVCF5, Space.VC, Jett McCandless, John Kim, Chris Nguyen, Alan Keating and Don Dodge.

Hubble's founders are no strangers to the wide world of Internet of Things (IoT) devices and services. Haro co-founded Life360, a location sharing and communication app for families, and saw the company through its listing on the Australian Securities Exchange in 2019. While working as Life360's chief technology officer, Haro said he was always looking to build useful hardware for families, too: fall detectors for elderly parents or GPS watches for kids. But the kind of network he was looking for to support these devices — one that had "low bandwidth, infrequent updates, but [...] globally accessible, very battery and cost efficient," as Haro described it — didn't exist.

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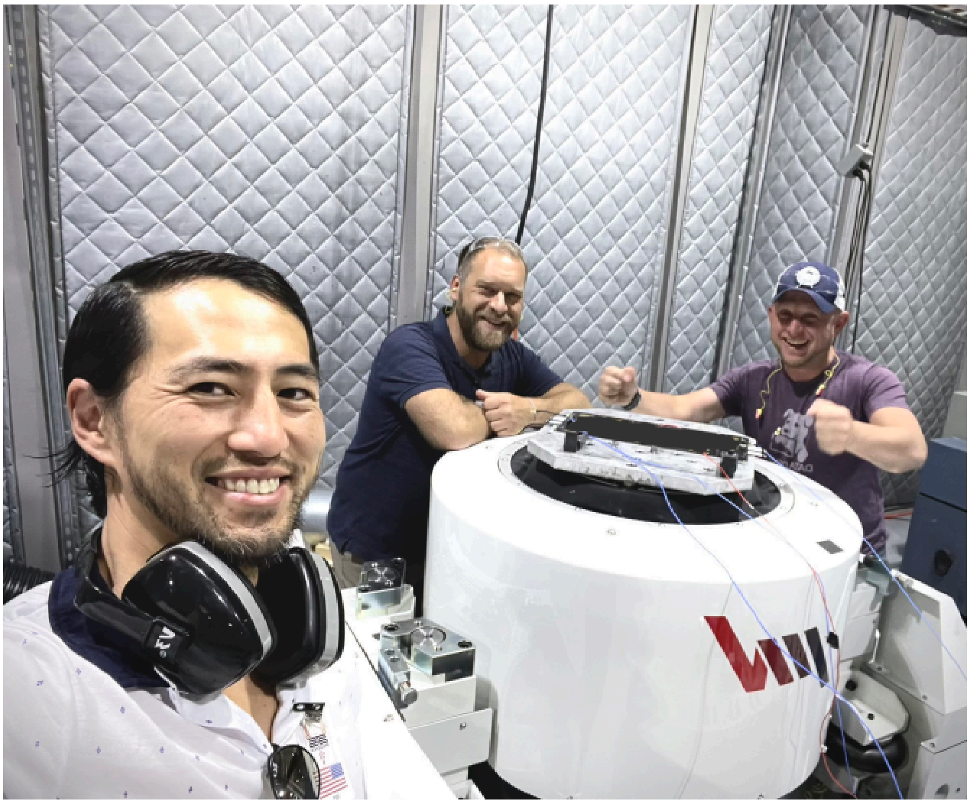
While mulling these issues, he met Ben Wild, who is now Hubble's CTO. Wild had previously founded Iotera, a company that was working on a crowd-sourced wireless network, and that was eventually sold to Ring (which was later bought by Amazon). Haro and Wild realized they could build the network they were looking for in space.

The pair brought in a third co-founder, aerospace engineer John Kim, whose career has spanned developing spacecraft systems at SSL, a subsidiary of Maxar Technologies, to aerospace consulting work. The trio officially incorporated Hubble Network in October 2021 and joined Y Combinator's Winter 2022 cohort.

"All three of us came together with this vision to connect any off the shelf Bluetooth chip directly to a satellite and really enable this network that will work anywhere in the world and be very battery and cost-efficient," Haro said. "We think that will unlock a whole bunch of really cool use cases."



JOHN KIM AND HUBBLE ENGINEERS WITH THE PAYLOAD AFTER IT PASSED THE FIRST ROUND OF QUALIFICATION TESTING.

IMAGE CREDITS: HUBBLE NETWORK

After launching four satellites next January, Hubble plans to build out its constellation to 68 satellites total over the next two-and-a-half years. While the first four satellites will provide global coverage on their own, Haro said that it will be about a six-hour gap until devices can update on the ground. Increasing the constellation to 68 birds means that a satellite will be overhead every 15 minutes or so — an update rate that is sufficient for “the vast majority” of customer use cases, Haro said.

While Hubble is clearly targeting existing Bluetooth devices — of which billions exist all over the world already — Haro is confident that the company’s network will solicit developers to build applications that don’t even exist yet.

“Eventually, if this network does exist, we think some of our biggest customers haven’t even incorporated yet because they haven’t been able to build these kinds of devices,” he said.

Topics: hubble network Space Transpose Platform

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